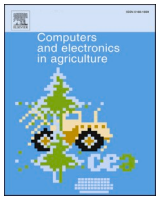
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Original papers



Sorted sample- wavelet feature clustering based on the range for classification of multiple nutrient deficiencies in tomato plants

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ABSTRACT

The phenomenon of physiological change in plants hyperpolarizes membrane potentials showing a significant change in electrophysiology. For plants to survive, develop, and reproduce, the proper ratio of nutrients is required. Malnourished plants exhibit unfavorable changes in their physiology. Nutrient deficiencies are one of the primary abiotic stresses occurred in plants which are commonly identified and classified through various hyperspectral imaging, and pattern recognition methods. In contrast to image-based methods, in this paper, the electrophysiological signal data of tomato plants in multiple nutrient deficiency states are classified. These electrical signals of tomato plants are acquired in healthy and in deficient states of calcium (Ca), manganese (Mn), nitrogen (N), and iron (Fe). The features are extracted by 7-level orthogonal wavelet decomposition using four different wavelets Daubechies (Db), Coiflet (Cf), Symlet (Sym), and Discrete Meyer (Dm). A novel range-based sorted sample clustering method is proposed to boost classification performance. Class assignment is performed in two modes homogeneous and non-homogeneous, where multi-class classification in non-homogeneous class assignment attained 99.4 % accuracy and in homogeneous class assignment a highest accuracy of 96.8 % is attained using the KNN classifier. Identifying nutrient deficiencies through electrophysiological signal analysis dominated the detection through imaging techniques, impacting a drastic change in crop protection practices.

1. Introduction

Many experts from various eras and civilizations firmly believe that plants possess far more complex powers than are typically apparent (Bose Sir, 1906; Stein, 1876). Monitoring the electrophysiology, also known as electrophytography (EPG) in some research, can be used to characterise the potentials with which plants pulsate, respond, and sense their environment. EPG signals play a crucial part in plant physiological processes, just like electrocardiogram (ECG) and electroencephalogram (EEG) signals do in biomedical applications (Debono & Souza, 2019; Simmi et al., 2020; Souza et al., 2017a). Studies analyzing Action potentials (AP), Resting potentials (RP), and Variation potentials (VP) while incorporating general parameters like frequency of occurrence, the potential, distance of propagation, and time series are being utilized to associate electrical signaling with physiological responses to environmental changes (Dziubińska et al., 2001; Pietruszka et al., 1997; Pyatygin et al., 2008). AP propagates along the length of the stem when the phloem is stimulated. VP is a slow wave and event-related potential occurred. Long-distance signaling is a systemic communication which is triggered by various types of stimuli or provoked by stresses.

The goal of systemic signalling is to activate response mechanisms in remote tissues by improving the ability of the whole plant to prepare its tissues for the upcoming challenge which provokes the change in electrophysiology (Sukhov et al., 2019). The measurement of low voltage electrical signals showed different environmental stimuli showed change in some characteristics of temporal dynamics of electrical signalling, including the level of complexity (Baluška, 2013; Souza et al., 2017a). If the stimulus or stress is continuous then non-linear dynamics of plant electrophysiology pushed them to critical state exhibiting spikes without a characteristic size indicating self-organized criticality (Souza et al., 2017a). The electrophysiological signals acquired during ideal conditions of plants differ in the case of plants undergoing stress.

The design of the experimental framework, which measures the plant's physiological response to stress, is significant. It is crucial to measure the electrophysiology or EPG signals both in a stress-free environment and under typical environmental conditions in order to

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assess the optimal physiological phenomena of a particular type of plant (Li et al., 2021; Rhodes & Nadolska-Orczyk, 2001). There are two types of stress where, abiotic stress is imposed by the environment and surroundings through non-living factors (Pande and Arora, 2017; Tuteja, 2007), while biotic stress is occurred by living organisms like bacteria, fungi, parasites, and harmful insects (Atkinson & Urwin, 2012; Behmann et al., 2015; Suzuki et al., 2014).

Nutrient deficiency is one of the main variables affecting crop productivity among all other factors. If a plant does not possess enough of a crucial nutrient needed for growth, nutrient deficiency results in. Plants will not grow effectively if they lack the necessary nutrients, and demonstrate a variety of symptoms to indicate the lack. For a healthy plant to be nourished and nurtured, macronutrients including calcium, magnesium, sulfur, nitrogen, and phosphorus are needed in greater amounts. Similarly, the compounds are generally referred to as micronutrients, copper, manganese, zinc, iron, and boron are some of the key nutrients that plants need in small amounts (Bottrill et al., 1970). In particular for the production of fruit and seeds, plants require nitrogen to promote rapid growth. It also improves the quantity and quality of leaves and speeds up plant maturation (Mills & Jones, 1979). Plant cell walls include calcium, which gives those walls their structural stability. It is stationary in plants and spends the entire growing season in the older tissue. Therefore, the younger leaves and leaf tips show the earliest signs of a deficit. For the assimilation of nitrogen, manganese activates the appropriate enzyme. The plant wilts and stops branching due to poor translocation of calcium and nitrogen ions, which are regarded as macronutrients. Plants require manganese for enzyme processes, respiration, and photosynthesis. Iron is necessary for plants in order to enhance the synthesis of chloroplast proteins and several enzymes. The youngest leaves are first affected by intervening chlorosis caused by micronutrients like iron and manganese. (Maia et al., 2019; Ruffel, n.d.; Watchareeruetai et al., 2018; Welch & Graham, 2004). Since it affects the natural ability of fruiting, flowering, and general plant growth, nutrient deficiencies must be identified and treated quickly.

The practice of signal processing techniques for retrieving hidden information from plant electrical signals has lagged behind the effort done for recording those signals. A variety of biomedical applications, including object identification, employ signal processing. While image processing has made significant progress in identifying plant diseases, there is still an essential limitation that disease identification using images is only possible after manifestation of visible symptoms. Through continuous signal monitoring, signal processing has demonstrated its ability to identify underlying diseases or stress. Additionally, digital signal processing methods like spectrograms and time–frequency analyses are essential for signal synchronisation and stress detection. The primary utilization signal processing methods was done in (Jingxia & Weimin, 2011) to comprehend the temporal and frequency properties of the recorded signal. Wavelet soft thresholding was implemented to analyze and de-noise electrical data from a jasmine tree (Wang & Li, 2010). Electrical signals from six distinct plant species were subjected to wavelet analysis in order to analyse their time–frequency properties and detect the presence of action potential (Lan-zhou et al., 2007). In order to evaluate the transmission speed of electrical signals collected from Aloe vera plants employing heat shock (burn), sulphuric acid, and irradiation, the lifting wavelet was utilized as a pre-processing step (Jingxia and Weimin, 2011). Moreover, investigators in (Yaddanapudia et al., 2017) attest to the existence of inter-plant communication in potted plants by claiming that an airborne signalling mechanism at a specific distance was demonstrated using electrophysiological signal analysis. In order to distinguish between mixed signals coming from the electrical signals of the epidermis cells, guard cells, and mesophyll cells, authors (Huang et al., 2010) employed independent component analysis ICA. They used light/darkness cycles as triggers to test the approach on generated signals and validate it with real surface-recorded electrical signals from plants. Based on the adaptable traits of the plants, a neural network with a Gaussian radial basis function was used to predict future electrical signal values. Several works attempted to classify various biotic and abiotic stresses, as well as short term stimuli in plants through their electrophysiological signals have attained a great scope to project this into an emerging field of research (Chatterjee et al., 2015, 2018; Debono & Souza, 2019; Najdenovska, Dutoit, Tran, Plummer, et al., 2021; Pereira et al., 2018; Sai et al., 2022a; Souza et al., 2017b; Tran et al., 2019; Tran & Camps, 2021). According to a report, such forecasting might be used in intelligent control systems, with potential applications in agriculture. In the previous study, the authors carried out an analysis by a familiar signal decomposition method termed empirical mode decomposition, and further nutrient deficiencies are binary classified using time-domain features (Sai et al., 2022b).

After a dedicated study of research background on electrical signaling in plants, the present study involves a design framework, data engineering, and classification of different types of nutrient deficiency classes. The present work contributes a signal decomposition methodology using orthogonal wavelets, where the signal is decomposed into different levels containing fine details in each level (say features). Then a sorted sample clustering method based on the range is proposed to separate potential ranges in each feature. Subsequently, a multi-class classification is performed to classify 5 classes of nutrient deficiencies (healthy, calcium, nitrogen, manganese, iron) in tomato plants by assigning classes in two different prototypes (homogeneous and non-homogeneous).

This entire methodology is organized into 4 different sections, where section 2 introduces the materials and methodologies proposed. Subsequent sections 3 and 4 follow a note on results achieved and comparative discussions. Finally, section 5 concludes the pros, cons, and further scope of this study.

2. Materials and methods

2.1. Database description

In the present study, databases from 15 tomato (Solanum lycopersicum) plants grown in a mixed substrate and specific nutrients from each 3 plants are removed for data acquisition and analyses purposes. Nutrient deficiencies of Fe, Ca, Mn, and N were considered for experiments, where all the experimentation and data acquisition is carried out at the Agroscope research station in Conthey, Switzerland. Initially, three plants were supplied with a nutrient solution and are considered healthy, while specific nutrients were removed from the nutrient solution and provided to a certain set of plants (set of 3 plants) (Sai et al., 2022b; Tran & Camps, 2021). Electrophysiological signals were acquired using an 8-channel- phytlsigns device from Vivent SA. Two insertion electrodes were positioned on the main stem and the leaf petiole to measure the potential difference between the electrodes for three weeks on each of the 15 plants. Powerline interference is shown in the thus recorded data, sampled at 500 Hz, by applying a notch filter at 50 Hz and 100 Hz. The signal's amplitude is expressed as a function of time and measured in millivolts (Tran & Camps, 2021).

The entire data acquired is made into 4 different chunks belonging to four different times of a day, as explained by authors (Sai et al., 2022b). The data in the present study is considered from data acquired during 10—11 am of the day. Fig. 1 portrays the signals acquired from tomato plants in healthy states and all nutrient deficiency states. Only 1000 samples of each signal are plotted against the magnitude (potentials) of the signal for a clear illustration of signal shape. This paper presented work on 18,00,000 samples of each plant signal.

In the vein, no two plants have the same potential charge concentration, even if they are grown in the same soil, at the same age, and in the same climate. Likewise, human vitals differ from human to human irrespective of age, height, and weight. The intake of nutrients and ion translocation determine the charge concentration or potentials of plants. The nutrients taken into focus in this investigation are listed in Table 1, together with experimental potential values obtained in both optimal

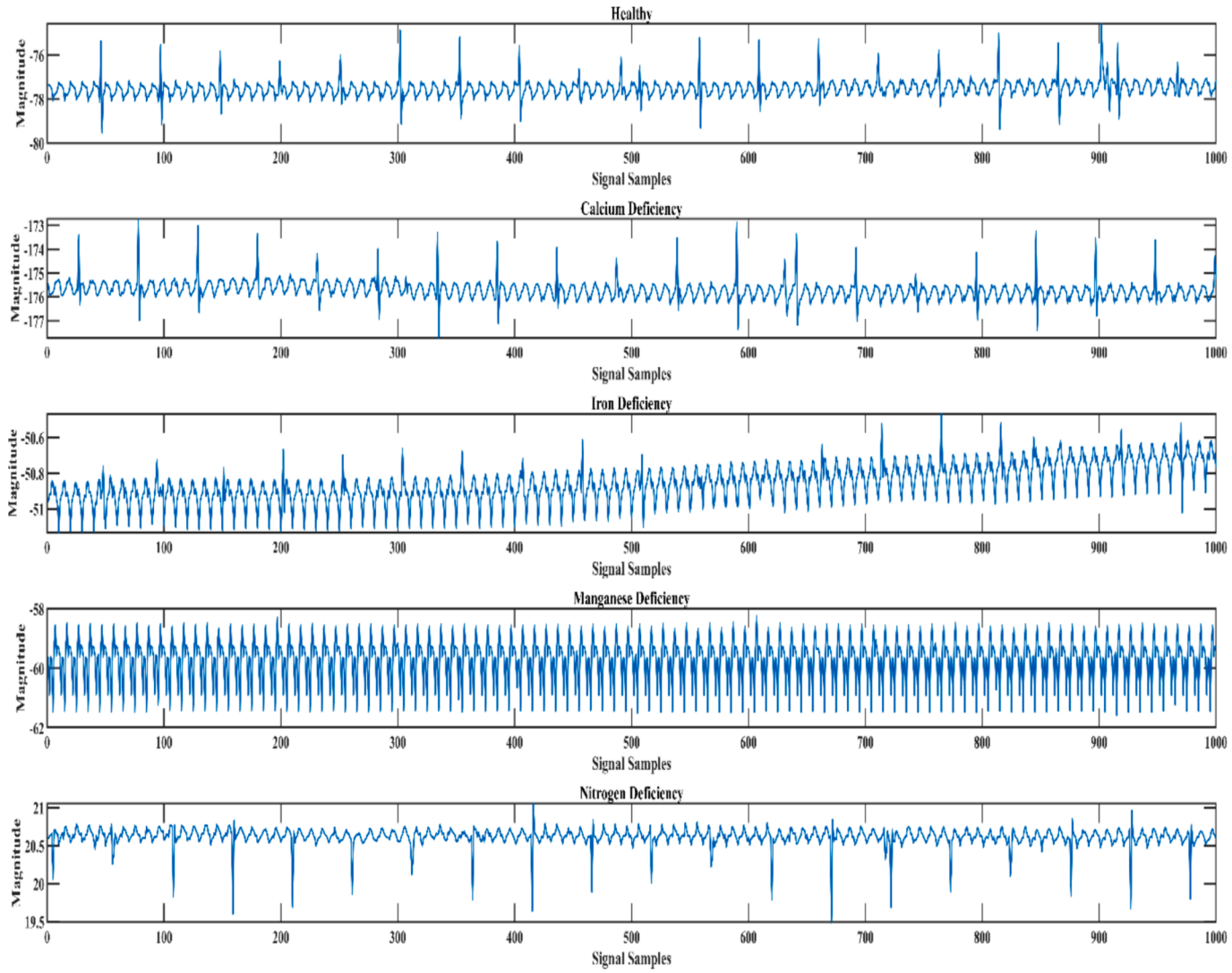


Fig. 1. Signal illustration in healthy and nutrient deficiency states.

Table 1
Electrophysiological potential information of tomato plants in healthy and deficient states.

Nutrient	Plant state	Plant 1			Plant 2			Plant 3		
		Min	**Max**	**K**	**Min**	**Max**	**K**	**Min**	**Max**	**K**
–	**Healthy**	−71.63	−91.32	−84.12	−18.25	−7.35	−12.55	−57.4	−30.12	−43.65
Ca	**Deficient**	−184.5	−158.6	−176.5	−36.68	−28.56	−32.45	−90.65	−77.52	−87.25
Fe	**Deficient**	−59.12	−44.82	−56.88	−23.7	−1.45	−11.76	−21.55	−12.58	−16.65
Mn	**Deficient**	−69.62	−58.56	−60.25	−160	106	98	−176.2	52.36	32.5
N	**Deficient**	12.6	28.8	18.58	−121.2	−66.5	−109.3	−71.29	−55.6	−58.5

and nutrient-deficient stress states.

Histograms of every plant are plotted to understand the range of potentials, where the minimum and maximum range of every plant in each healthy and nutrient-deficient state are portrayed in Table 1. Also, the median value of the entire signal where the potential value which is having the highest number of repetitions is noted as K. The value of K for every plant used in this investigation is portrayed in Table 1. The findings in Table 1 demonstrated that in plant 1, the ion concentration is −71.63 in a normal state and rises to −184 in a situation of calcium deficiency. Similar outcomes for other nutrients indicate the specific type and potentials demonstrating variability in normal healthy states and during deficiency states. Presenting the variations in these potentials at regular intervals could offer a thorough examination to help comprehend nutritional deficits that are still in their early stages. Further approaches could be established through electrophysiological analysis of coupled nutrient deficit stressors. Nevertheless, samples in this study are clustered according to the range of potentials that vary in entire the signal.

2.2. *Work methodology*

The entire workflow of this study is illustrated in Fig. 2. A stepwise pragmatic methodology is being followed to analyze and classify multiple stresses.

Initially, signal data is prepared by checking for any necessity for artifact removal. This data is processed to decompose using orthogonal

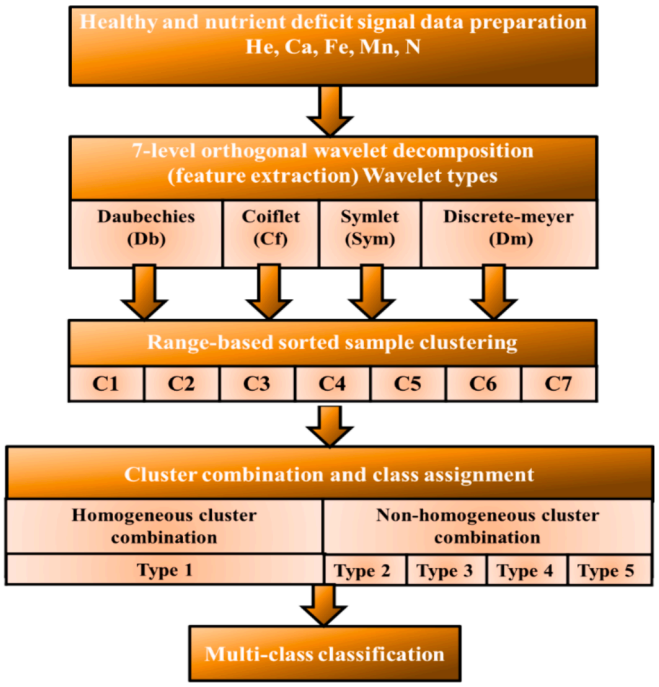


Fig. 2. A step-by-step procedure of the entire methodology illustrated as a flow diagram.

wavelets. The filter coefficients of orthogonal wavelets are usually used to define them. The quadrature relation between the two filters can be used to get the filter coefficients. The capacity to concurrently determine a signal's time, position, and frequency is the primary benefit of wavelets. Thus, wavelet decomposition is the method of choice in this study for splitting the signal into modest features. The features extracted through wavelet decomposition, are clustered in different groups based on ranges. This range-based clustering is proposed to replace the normalisation of data, since data grouping fits best instead of data manipulation. Then, multiple classes are assigned by grouping clusters in homogeneous and non-homogeneous forms. Class prediction became clearer and easier with the use of cluster-based class assignments, which enhanced classifier performance. Finally, multi-class classification is performed.

2.2.1. *Wavelet decomposition*

The multiscale signal processing techniques integrated wavelet decompositions as a major part. They are put into practice by employing cascaded filter banks, where the lowpass and high-pass filters adhere to predetermined rules (Gaouda et al., 1999; Heijmans & Goutsias, 2000; Sairamya et al., 2019). A better knowledge of wavelet decompositions and their function in vision is provided by works in various computational mathematics. Their novelty links with traditional signal processing principles that offer a deeper functional understanding of various applications (Akansu et al., 2010; Graps, n.d.). The signal is divided into a number of sub-signals (or sub-bands) through wavelet decomposition, each of which also represents information at various scales. This study uses four different wavelets to perform decomposition (Haddad et al., 1993). The wavelets used are Db, Sym,Cf, and Dmey as shown in Fig. 3.

Algorithm:

The discrete wavelet transform consists of no more than log2 L steps for a signal m of length L. Initiating with k, the first stage generates two sets of coefficients: coA1 for approximation and coD1 for detail (Tahir Ismail, n.d.). The approximation and detail coefficients are obtained by convoluting signal s with the lowpass filter Lo and the highpass filter Hi, subsequently performing collaborative decimation termed as down sampling. Two even-length real-valued vectors are used to specify

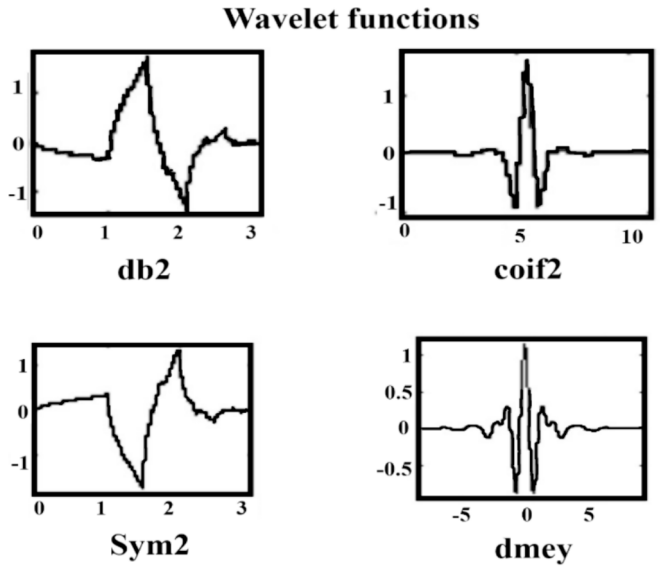


Fig. 3. Different types of wavelet functions.

wavelet decomposition filters. The lowpass and high pass decomposition filters are designated as Lo and Hi, respectively. Lo and Hi's lengths are required to precisely be equal. Each filter has a length equal to 2n. The signal m is convolved with filter and downsampled to keep only even indexed elements. The respective lengths of signals and coefficients are shown in equations (2.1), (2.2), and (2.3).

$$\textit{Length of signal } m \textit{ is noted as } L = \textit{length}(m) \tag{2.1}$$

$$\textit{Length of signals } E \textit{ and } F = L + 2n - 1 \tag{2.2}$$

$$\textit{Length of coefficients } coA1 \textit{ and } coD1 = (L - 1/2) + n \tag{2.3}$$

This process of obtaining approximation and detailed coefficients using discrete wavelet transform is illustrated in Fig. 4. Using the same strategy demonstrated in Fig. 4, the further approximation coefficients are generated from coA1 as coA2 and coD2, and so on.

The wavelet decomposition vector's coefficients are divided by level using the decomposed signal, which is represented as a vector. The length of the original signal and the number of coefficients by level are both represented in the length vector. The wavelet decomposition vector's coefficients are level-parsed using this method. This level-7 decomposition is illustrated in Fig. 5, by demonstrating the way the decomposition vector and length vector are arranged.

In this study, the signals obtained in different nutrient deficiency and healthy states are processed to decompose using four different wavelets. The respective decomposition coefficients are used as features. These features are clustered together in different groups to attain better classification performance. This clustering method is explained in detail in

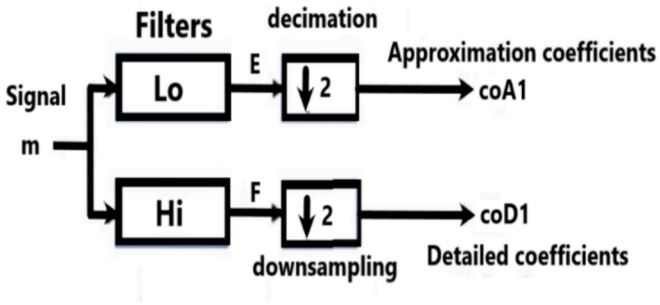


Fig. 4. Illustration of decomposition process using filters and down sampling.

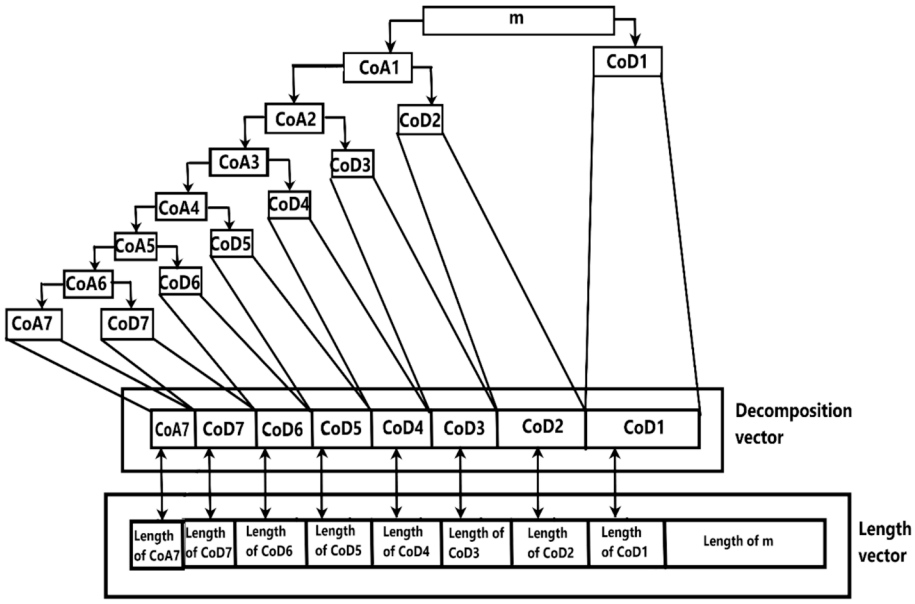


Fig. 5. Illustration describing 7-level orthogonal wavelet decomposition.

the next section.

2.2.2. Proposed technique (Range –based sorted sample clustering method)

The wavelet decomposition decomposed signal components in different signal scales. Although the attained decomposition has enough information in different bands, the nutrient deficiencies in EPG signals show a very less potential separability to attain the attention of classifier. So there comes a necessity to propose a technique to engineer the data for attaining better classification performance. In this study, the data is clustered and classes are assigned in different types. The 7 decomposed vectors obtained after decomposition in 7 level decomposition are considered as 7 different features. The protocol for dividing C1 to C7 is the decomposed signal using wavelet decomposition parted the signal into 7 different levels by separating fine details of the signal in each level, which indeed offers a simultaneous localization in time and frequency domain. This wavelet decomposition is performed on each signal in healthy and deficient states individually. Wavelet decomposition is performed using MATLAB, libraries. These features in each level are named after F1, F2, F3, F4, F5, F6, and F7 as illustrated in part (a) of Fig. 6. Initially, the features values or samples are sorted in ascending order (potential information). Then, if the total length of each decomposed vector or feature is said to be N, then each feature is divided

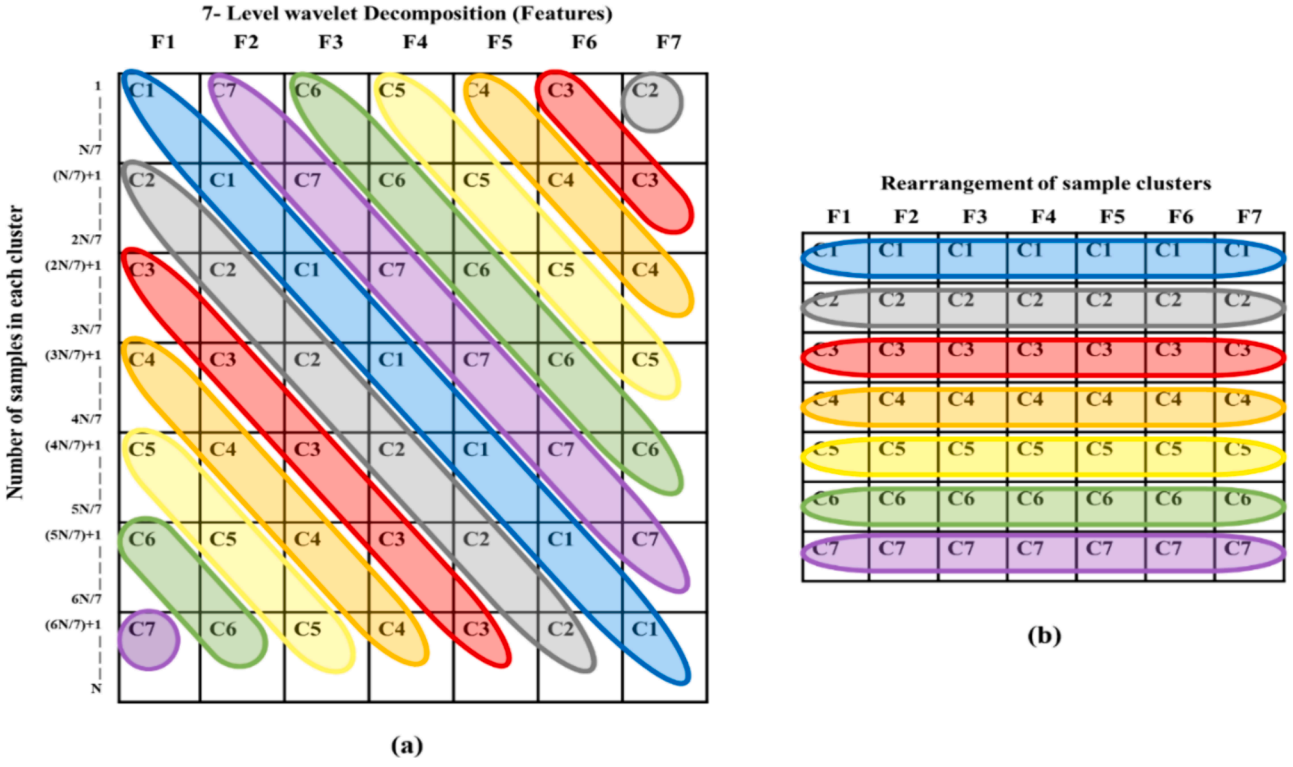


Fig. 6. (a) Range based sorted sample clustering illustration (b) Rearranged sample clusters after clustering.

into 7 clusters, where length of each cluster is N/7. The number of samples or feature values in each cluster is shown in Fig. 6 (a).

The cluster C1 in feature F1 contains feature values from 1 to N/7 but the the from feature F2, (N/7) + 1 to 2 N/7 values descent into cluster C1. Similarly, the feature values from (2 N/7) + 1 to 3 N/7 descent into the cluster C1. Likewise, the cluster groups formed from every feature as shown in part (a) of Fig. 6, are arranged in the form as shown in Fig. 6 (b). The data seperability of feature values in each cluster after clustering is shown in the Fig. 7 through histogramic representation.

2.2.3. Class assignment

The range-based samples are sorted and clustered for every class of nutrient deficiency and healthy class. In this article, 5 classes are considered to classify, where healthy class is denoted by class 0, Ca deficiency by class 1, Fe deficiency by class 2, Mn deficiency by class 3, and Ca deficiency by class 1. Classes are assigned in two types- homogeneous and non-homogeneous. Homogenous type of class assignment consists of assigning same clusters from every class, while non– homogeneous class assignment comprises different clusters from every class. Class separability is the major parameter for the approach of learning from data and making new observations based on assumptions. If the learning data is clumsy and repetitive the classifier obscures between classes and assumptions might be manipulated. On the training data, a separability analysis can be performed to calculate the predicted error in the classification for different feature combinations. In this article, surface plots are used to display three-dimensional data of non-homogeneous cluster combinations. Fig. 8 shows an illustration of cluster assignment for every class while homogeneous assignment for every cluster is noted as type 1 in whole study and non-homogeneous assignment is performed in four types of cluster combinations.

2.2.4. Multi-class classification

A multi-class classification scheme is adopted to classify 7 features by the combination of various nutrient deficiency cases. The principle of multi-class classification is to classify entire data into the required number of groups (five) based on the classification rule. Classes are assigned to respective clusters and trained using supervised machine learning methods. This study uses K- nearest neighbor (KNN) classifier to classify the data. KNN is a supervised classifier built on the lazy learner algorithm, a non-parametric classification approach. KNN does not use training data to learn; instead, it stores the full database and uses similarity tests to establish classes. An algorithm searches for the K-number of nearest neighbors using distance metrics (Mahalonabis, chebychev, Minkowski, etc.) to categorize class values (Begum et al., 2015; Guo et al., 2003). The value of K is first assigned (number of neighbors). The KNN classifier parameters in this study are noted as the number of neighbours-10, distance metric – mahalonabis, and distance weight – equal. While assigning a value to any new observation in KNN, the number of neighbors are considered by determining the appropriate k value. The model overfits on the training set, resulting in a significant error rate on the validation set, for very low values of k = 1,2,5 etc. Conversely, the model underperforms on the train and validation sets for high values of k = 20,50,70. So, Results in this manuscript are attained by adjusting the parameters according to the dataset neither to overfit nor underfit. As earlier stated, the k value 10 best fits this database. The classification results achieved to classify multiple nutrients are discussed in the next section.

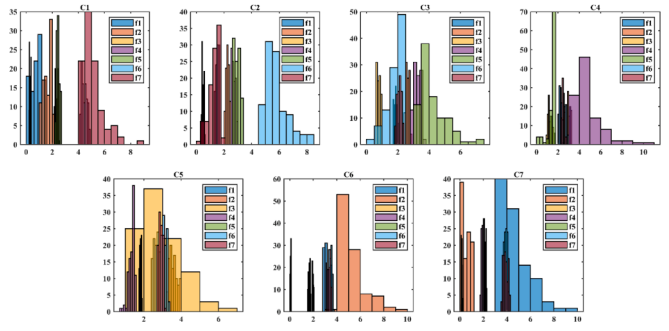


Fig. 7. The histogram representation of each cluster showing data separability of each feature per cluster.

3. Results

The classification of nutrient deficiency in signal data was examined in conjunction with observations of decomposed signals, which indicated that it might perform substantially when grouped into different clusters. The methods used to evaluate signals and execute data engineering for improved classification performance with respect to data changes, the entire work in this study followed a step-by-step process, which was explained in earlier sections.

The resultant data separability attained in each cluster after performing the proposed technique of clustering has widely increased as already shown in Fig. 6. The most fundamental visual representations of three-dimensional data are surface plots, which display a functional relationship among a designated dependent variable and two independent variables rather than considering individual data points. In this study, potential and sample distribution are considered as independent variables and class is the dependent variable. The surface graph representing cluster combinations of type 2, 3 4, and 5 are shown in Fig. 9. In this figure, it is visualized that the range-based sorted sample clustering technique together with a non-homogeneous class assignment has attained the highest-class separability which enhances the classification performance. The three-dimensional surface graphical representation illustrates the heights above the grid are potential values (z-axis) represented in various compact collars for every class above the x-y plane, while the color of the surface varies according to the height of potential. The distribution of samples and class separability is illustrated in the x-y plane of the graph, where the heights of potentials varied in every class portraying the best separability in all four types of cluster combinations. Observations clearly illustrated the height of every class with either a positive or negative slope created a separable hyperplane distinguishing the classes in all types of cluster combinations, which is a greater achievement to enhance the classification performance. This data is classified using KNN supervised algorithm as a classifier while the reported results attained a classification rate up to 96 % in type 1 cluster combinations. The reported results attained by each cluster in the Type 1 combination in the case of every wavelet are shown in Table 2. The Cf wavelet has attained a classification rate greater than 96 % in the clusters C1, C4, and C5. These clusters provided good performance, while the work can be extended by grouping these three clusters to understand the classifier performance. Also, the clusters C2, C3, C6 and C7 in case of Cf wavelet decomposed features have attained classification rate greater than 95 %. Overall, observations from Type 1 homogenous cluster combinations depict that Cf wavelet decomposition features attained best performance when compared to other wavelets. Secondarily the Dm wavelet achieved good accuracy up to 95.8 % in case of C2 cluster combination, while the other clusters depicted classification rate above 94 %. Also, the accuracies attained by Db and Sym wavelets are greater than 90 %, which is not a coarsest classification rate. The performance metrics depicted by non-homogeneous cluster combinations of type 2, 3, 4, and 5 and an average classification rate of homogeneous clusters of type 1 are shown in Table 3.

Observations show that all the non-homogeneous clusters attained accuracies greater than 95 % while Cf and Dm wavelets attained classification rate greater than 99 %. Type 4 and Type 5 cluster combinations as shown in Fig. 7, depicted classification rate as 99.2 % and 99.4 %. Overall, in both homogeneous, and non-homogeneous cluster combinations, the wavelets Cf and Dm portrayed the highest performance metrics, while non-homogeneous clusters are better than homogeneous

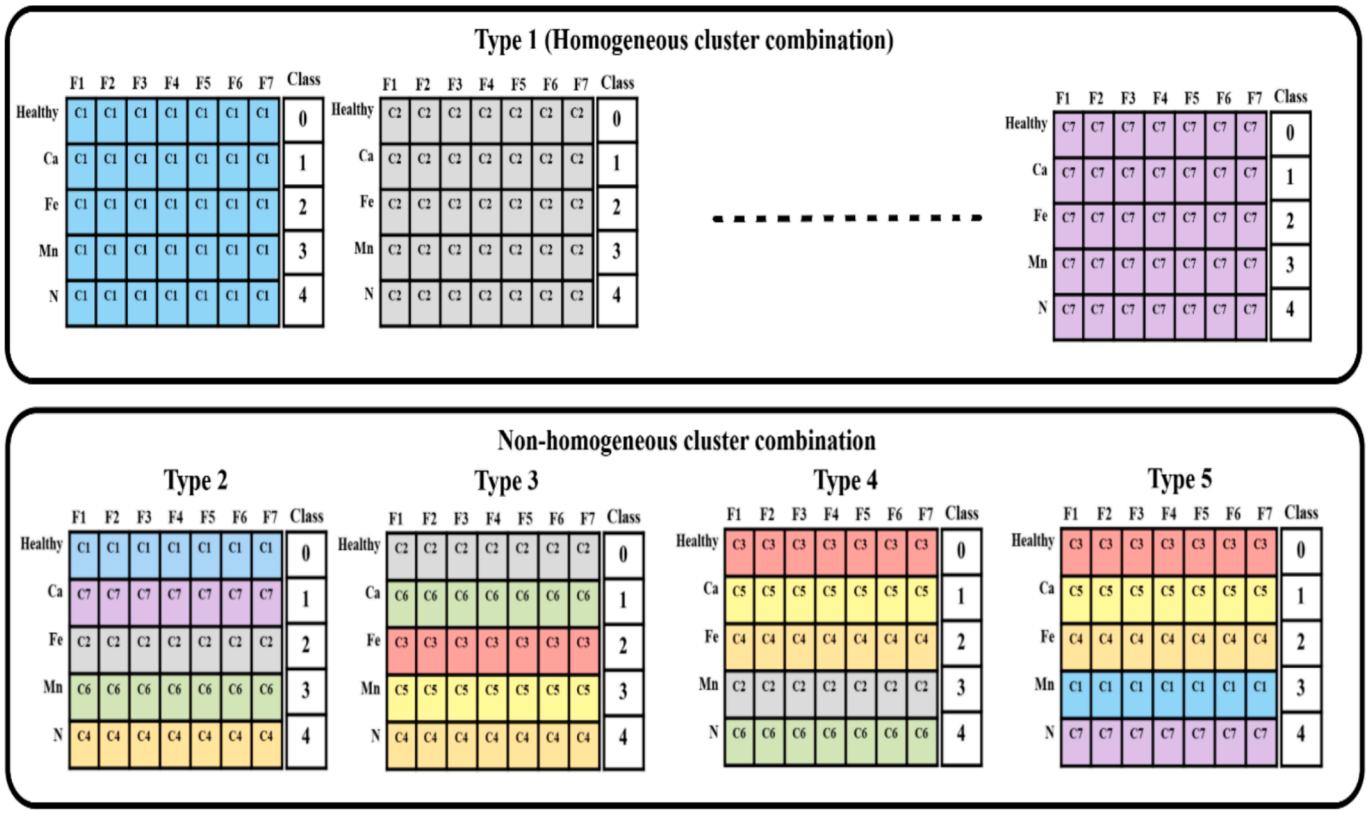


Fig. 8. Cluster combination and class assignment illustration.

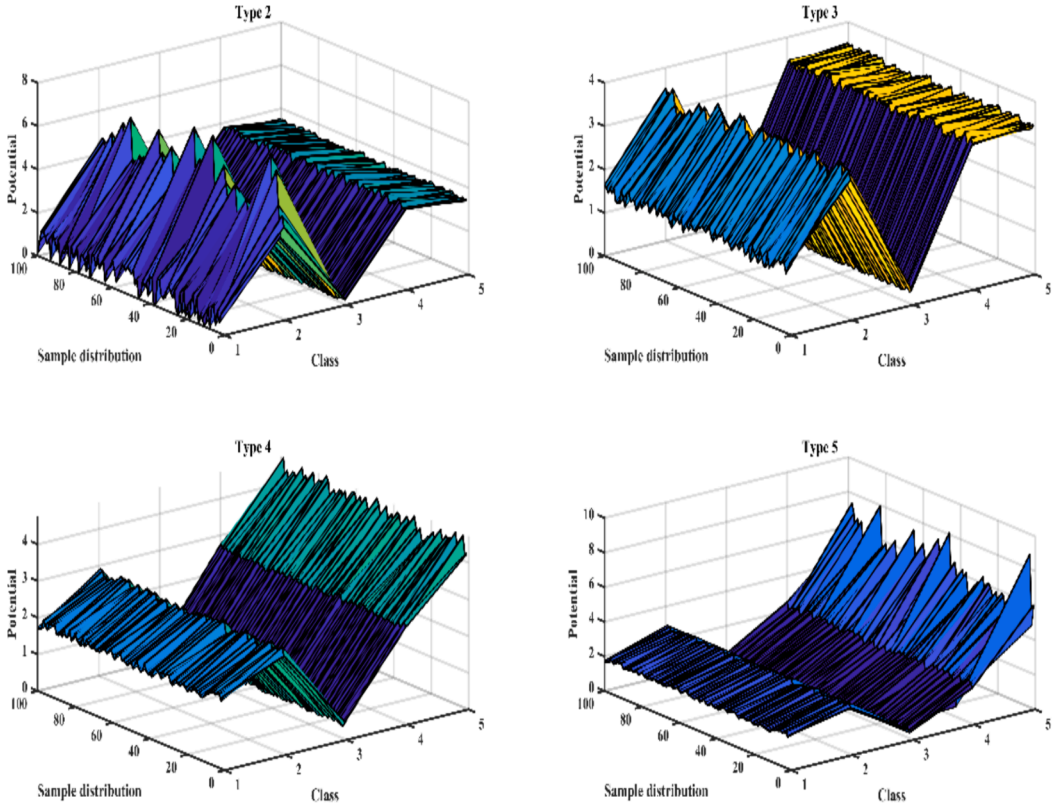


Fig. 9. Surface plot visualization of data in three-dimensional representation for all non-homogenous cluster combinations.

Table 2
The classification rates attained by each wavelet to classify various nutrient deficiencies in homogeneous cluster combinations.

Type 1	Daubechies (Db)	Coiflet (Cf)	Symlet (Sym)	Discrete meyer (Dm)
C1	92.2	96.8	91.2	94.8
C2	93.4	95.7	92.2	95.8
C3	90.8	95.8	90.6	94.5
C4	91.9	96.4	93.2	94.6
C5	94.4	96.8	93.8	94.9
C6	90.6	95.6	90.9	94.3
C7	92.7	95.5	93.2	94.8

Table 3
The classification rates attained by each wavelet to classify various nutrient deficiencies in non-homogeneous and homogeneous (average of Type 1) cluster combinations.

Combinations	Daubechies (Db)	Coiflet (Cf)	Symlet (Sym)	Discrete meyer (Dm)
Type 1	92.2	96	92.1	94.8
Type 2	96.9	97.8	95.6	97.5
Type 3	97.4	98.9	96.8	98.7
Type 4	95.8	**99.2**	95	**99.2**
Type 5	96.8	98.5	95.8	**99.4**

cluster classes.

It is enlightening that wavelet decomposition enabled the rapid building of a database with different types of wavelets, data clustering, and cluster combinations, which indeed resulted in 99 % of classification performance accomplishment for evaluating minute differences in the potentials of multiple deficiencies at a time. These discoveries may be important for understanding the plant metabolism discussed in the next section.

4. Discussion

The manufacturing and service sectors of the Indian economy are based on agriculture, which is its backbone. A considerable decline in both the quality and quantity of agricultural goods is brought on by plant nutrient deficiencies. Two essential elements in agricultural output are early deficiency symptom detection and precise classification. Due to ineffective nutrient deficiency detection and erroneous classification, signal acquisition and processing methods have largely superseded image-based detection methods. The image processing methods of deficiency detection and classification include existing imaging techniques followed by detection and classification using machine learning and deep learning methods. Many researchers attempted to build various computational models to classify the nutrient deficiencies in various plant species and attained a good classification rate, but most of them followed image processing techniques. The studies based on signal processing are very few and are still in the initial stages of study, where some studies are shown in Table 4.

Similar to human beings, the pulse rate or respiration rate varies with age, physical activity, dietary habits, etc. Similarly, not all plants have the same ion concentration, even though they are all grown in the same medium, under the same climatic conditions, and are identical in age. The charge concentration or potentials of plants depend on the uptake of nutrition and ion translocation. Observations of attained plant potentials in this study illustrate this point, where this became a motivation to analyze signals in the time–frequency domain to reduce information loss. Wavelet decomposition in general has more applications in biomedical signal processing, while it competed to make an equal contribution to analyze plant signals correspondingly. In this study, wavelet decomposition attained a 99.4 % classification rate the highest using Dm wavelet in type 4 cluster combination. The line graph visualization of different wavelet performances in each cluster combination is shown in Fig. 10. Overall, Cf and Dm performed best in all cases of classification comparatively.

The studies using electrophysiological signals in plants can be elaborated in various domains, likewise, the latest findings that data demonstrating variations in tomato fruit electrical activity complexity as they ripen and explain potential physiological processes involved. The signals' relative complexity, as determined by entropy, changed as the fruit ripened and thereby, resulted in the data revealing a reduction in signal complexity in the breaker stage, perhaps as a result of some physiological mechanism that eventually takes precedence at the expense of others (Reissig et al., 2023). Also, many researchers performed analyses to classify various abiotic, and biotic stresses through time series regression, time domain analyses, spectral and statistical feature extraction, frequency domain analyses, filter design, time–frequency domain, and time series analyses (Chatterjee, 2017; Chatterjee et al., 2014, 2015, 2017, 2018; Das et al., 2015; Gurovich & Hermosilla, 2009; Najdenovska, Dutoit, Tran, Plummer, et al., 2021; Najdenovska, Dutoit, Tran, Rochat, et al., 2021; Pereira et al., 2018; Sai et al., 2022a; Souza et al., 2017a; Tran et al., 2019). All these studies include plant electrical signals to detect and classify stress. Studies linking abiotic and biotic stress detection using image processing are thousands in number, while signal studies not even lie in the class of hundreds in the past decade. There is a huge necessity to improve studies on plant electrical signals.

Although, the categorization of plant illnesses using various Deep Learning (DL) architectures attained higher classification rates the

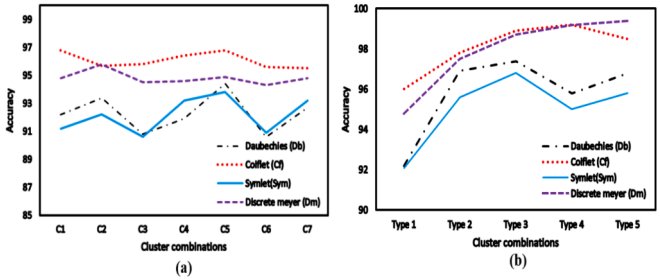


Fig. 10. (a) Line graph illustration of various wavelet decomposed features in **homogeneous** cluster assignment (b) Line graph illustration of various wavelet decomposed features in **non-homogeneous** cluster assignment.

Table 4
A note on the latest studies relevant to nutrient deficiency detection and classification.

Author	Nutrient deficiency	Species	Domain of analysis	Classifier	Performance
(Ibrahim et al., 2022)	N, K, Mg, B, Zn, Mn	Palm	Image processing	CNN	94.26 %
(Jose et al., 2021)	Ca, Mg, N, P, K, S	Tomato	Image processing	ANN	88 %
(Sabri et al., 2020)	Mg, N, K		Image processing	Random forest	78.35 %
(Sathyavani et al., 2022)	Mg, Ca, P, N, S, K	Rice	Image processing	Dense net-BC	96 %
(Kumar et al., 2022)	4 major nutrients	Apple	Image processing	Deep-CNN	98.25 %
(Tran & Camps, 2021)	Fe	Tomato	Signal processing	GBT	76 %
(Sai et al., 2022b)	Ca, Mn, Fe, N	Tomato	Signal processing	KNB, GNB, EBT, QDA	98.5 %
Present study	Ca, Mn, Fe, N	Tomato	Signal processing	KNN	99.4 %

major limitation of imaging techniques exists where the deficiency can be detected and classified only after visual symptom appearance, while in signals not only nutrient deficiencies various abiotic and biotic stresses can be identified before the appearance of symptoms. Acquisition of plant signals continuously in large-scale production systems to perform more research in this area might result in more accurate tracking and classification of plant diseases. Agriculture and cultivation in correspondence with the knowledge of the processes that take place in plant physiological phenomena have benefited considerably from many biological studies, but the application of plant electrophysiology linked with biological study has enormous potential for increasingly precise and sustainable crop protection. Incorporating automated data acquisition systems, in correspondence with signal analysis, alongside artificial intelligence techniques in appropriate proportions distributes knowledge to more accurately classify various plant deficiencies during early stages prior to symptom appearance, which causes a significant change in the agricultural sector to control plant growth and enhance plant protection techniques.

5. Conclusion

The time–frequency domain analysis together with machine learning algorithms portrayed a good classification performance to classify multiple nutrient deficiencies in a single stretch. Wavelet decomposition played a crucial role in feature extraction for classification. The signals are decomposed into 7 seven levels since either a lower or higher number of features might complicate the task. With the assessment of feature matrices obtained and their impact on the classification process, a novel sample clustering method is proposed which influenced the classification performance to a greater extent. The class assignment is performed in two different methods while non-homogeneous cluster combinations attained a multi-class classification rate of 99.4 %. The previous studies reported on the same databases have attained 76 % and 98.5 % accuracies, while they depicted a binary classification. Optimal and restorative crop protection practices are greatly aided by accompanying advancements in techniques and analyses to evaluate plant status prior to the onset of symptoms. Designing frameworks and developing computational algorithms to assess real-time data, identify the type of deficiency a crop is experiencing, and determine its state of stress is indeed required.

CRediT authorship contribution statement

Kavya Sai: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Data curation, Conceptualization. **Neetu Sood:** Supervision, Project administration, Formal analysis. **Indu Saini:** Supervision, Project administration, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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